

Университет ИТМО

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Факультет Программной Инженерии и Компьютерной техники

Курсовая работа №2

По дискретной математике

Вариант №19

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## Задание

Построить комбинационную схему реализующую функцию  $C = (A - 1)_{\text{mod}57}$ , где  $A = (a_1, a_2, a_3, a_4, a_5, a_6)$ ,  $C = (c_1, c_2, c_3, c_4, c_5, c_6)$

## Таблица истинности

| №  | a <sub>1</sub> | a <sub>2</sub> | a <sub>3</sub> | a <sub>4</sub> | a <sub>5</sub> | a <sub>6</sub> | c <sub>1</sub> | c <sub>2</sub> | c <sub>3</sub> | c <sub>4</sub> | c <sub>5</sub> | c <sub>6</sub> |
|----|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|
| 0  | 0              | 0              | 0              | 0              | 0              | 0              | 0              | 0              | 0              | 1              | 0              | 1              |
| 1  | 0              | 0              | 0              | 0              | 0              | 1              | 0              | 0              | 0              | 0              | 0              | 0              |
| 2  | 0              | 0              | 0              | 0              | 1              | 0              | 0              | 0              | 0              | 0              | 0              | 1              |
| 3  | 0              | 0              | 0              | 0              | 1              | 1              | 0              | 0              | 0              | 0              | 1              | 0              |
| 4  | 0              | 0              | 0              | 1              | 0              | 0              | 0              | 0              | 0              | 0              | 1              | 1              |
| 5  | 0              | 0              | 0              | 1              | 0              | 1              | 0              | 0              | 0              | 1              | 0              | 0              |
| 6  | 0              | 0              | 0              | 1              | 1              | 0              | 0              | 0              | 0              | 1              | 0              | 1              |
| 7  | 0              | 0              | 0              | 1              | 1              | 1              | 0              | 0              | 0              | 1              | 1              | 0              |
| 8  | 0              | 0              | 1              | 0              | 0              | 0              | 0              | 0              | 0              | 1              | 1              | 1              |
| 9  | 0              | 0              | 1              | 0              | 0              | 1              | 0              | 0              | 1              | 0              | 0              | 0              |
| 10 | 0              | 0              | 1              | 0              | 1              | 0              | 0              | 0              | 1              | 0              | 0              | 1              |
| 11 | 0              | 0              | 1              | 0              | 1              | 1              | 0              | 0              | 1              | 0              | 1              | 0              |
| 12 | 0              | 0              | 1              | 1              | 0              | 0              | 0              | 0              | 1              | 0              | 1              | 1              |
| 13 | 0              | 0              | 1              | 1              | 0              | 1              | 0              | 0              | 1              | 1              | 0              | 0              |
| 14 | 0              | 0              | 1              | 1              | 1              | 0              | 0              | 0              | 1              | 1              | 0              | 1              |
| 15 | 0              | 0              | 1              | 1              | 1              | 1              | 0              | 0              | 1              | 1              | 1              | 0              |
| 16 | 0              | 1              | 0              | 0              | 0              | 0              | 0              | 0              | 1              | 1              | 1              | 1              |
| 17 | 0              | 1              | 0              | 0              | 0              | 1              | 0              | 1              | 0              | 0              | 0              | 0              |
| 18 | 0              | 1              | 0              | 0              | 1              | 0              | 0              | 1              | 0              | 0              | 0              | 1              |
| 19 | 0              | 1              | 0              | 0              | 1              | 1              | 0              | 1              | 0              | 0              | 1              | 0              |
| 20 | 0              | 1              | 0              | 1              | 0              | 0              | 0              | 1              | 0              | 0              | 1              | 1              |
| 21 | 0              | 1              | 0              | 1              | 0              | 1              | 0              | 1              | 0              | 1              | 0              | 0              |
| 22 | 0              | 1              | 0              | 1              | 1              | 0              | 0              | 1              | 0              | 1              | 0              | 1              |
| 23 | 0              | 1              | 0              | 1              | 1              | 1              | 0              | 1              | 0              | 1              | 1              | 0              |
| 24 | 0              | 1              | 1              | 0              | 0              | 0              | 0              | 1              | 0              | 1              | 1              | 1              |
| 25 | 0              | 1              | 1              | 0              | 0              | 1              | 0              | 1              | 1              | 0              | 0              | 0              |
| 26 | 0              | 1              | 1              | 0              | 1              | 0              | 0              | 1              | 1              | 0              | 0              | 1              |
| 27 | 0              | 1              | 1              | 0              | 1              | 1              | 0              | 1              | 1              | 0              | 1              | 0              |
| 28 | 0              | 1              | 1              | 1              | 0              | 0              | 0              | 1              | 1              | 0              | 1              | 1              |
| 29 | 0              | 1              | 1              | 1              | 0              | 1              | 0              | 1              | 1              | 1              | 0              | 0              |
| 30 | 0              | 1              | 1              | 1              | 1              | 0              | 0              | 1              | 1              | 1              | 0              | 1              |
| 31 | 0              | 1              | 1              | 1              | 1              | 1              | 0              | 1              | 1              | 1              | 1              | 0              |
| 32 | 1              | 0              | 0              | 0              | 0              | 0              | 0              | 1              | 1              | 1              | 1              | 1              |
| 33 | 1              | 0              | 0              | 0              | 0              | 1              | 1              | 0              | 0              | 0              | 0              | 0              |
| 34 | 1              | 0              | 0              | 0              | 1              | 0              | 1              | 0              | 0              | 0              | 0              | 1              |
| 35 | 1              | 0              | 0              | 0              | 1              | 1              | 1              | 0              | 0              | 0              | 1              | 0              |
| 36 | 1              | 0              | 0              | 1              | 0              | 0              | 1              | 0              | 0              | 0              | 1              | 1              |
| 37 | 1              | 0              | 0              | 1              | 0              | 1              | 1              | 0              | 0              | 1              | 0              | 0              |
| 38 | 1              | 0              | 0              | 1              | 1              | 0              | 1              | 0              | 0              | 1              | 0              | 1              |
| 39 | 1              | 0              | 0              | 1              | 1              | 1              | 1              | 0              | 0              | 1              | 1              | 0              |
| 40 | 1              | 0              | 1              | 0              | 0              | 0              | 1              | 0              | 0              | 1              | 1              | 1              |
| 41 | 1              | 0              | 1              | 0              | 0              | 1              | 1              | 0              | 1              | 0              | 0              | 0              |

↙ Переполнение

|    |   |   |   |   |   |   |   |   |   |   |   |   |
|----|---|---|---|---|---|---|---|---|---|---|---|---|
| 42 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 0 | 1 |
| 43 | 1 | 0 | 1 | 0 | 1 | 1 | 1 | 0 | 1 | 0 | 1 | 0 |
| 44 | 1 | 0 | 1 | 1 | 0 | 0 | 0 | 1 | 0 | 1 | 1 | 1 |
| 45 | 1 | 0 | 1 | 1 | 0 | 1 | 1 | 0 | 1 | 0 | 0 | 0 |
| 46 | 1 | 0 | 1 | 1 | 1 | 0 | 1 | 0 | 1 | 1 | 0 | 1 |
| 47 | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 1 | 0 |
| 48 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 1 | 1 |
| 49 | 1 | 1 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 0 | 0 | 0 |
| 50 | 1 | 1 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 1 |
| 51 | 1 | 1 | 0 | 0 | 1 | 1 | 1 | 0 | 0 | 1 | 0 | 0 |
| 52 | 1 | 1 | 0 | 1 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 1 |
| 53 | 1 | 1 | 0 | 1 | 0 | 1 | 1 | 0 | 1 | 0 | 0 | 0 |
| 54 | 1 | 1 | 0 | 1 | 1 | 0 | 1 | 0 | 1 | 0 | 0 | 1 |
| 55 | 1 | 1 | 0 | 1 | 1 | 1 | 1 | 0 | 1 | 1 | 0 | 0 |
| 56 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 1 | 1 |
| 57 | 1 | 1 | 1 | 0 | 0 | 1 | 1 | 1 | 0 | 0 | 0 | 0 |
| 58 | 1 | 1 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 59 | 1 | 1 | 1 | 0 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 1 |
| 60 | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 |
| 61 | 1 | 1 | 1 | 1 | 0 | 1 | 1 | 0 | 0 | 0 | 1 | 1 |
| 62 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 |
| 63 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 1 | 0 | 1 |

## Минимизация булевых функций на картах Карно

$a_5a_6$

|             |    |    |    |    |
|-------------|----|----|----|----|
|             | 00 | 01 | 11 | 10 |
| $a_3a_4$ 00 |    |    |    |    |
| 01          |    |    |    |    |
| 11          |    |    |    |    |
| 10          |    |    |    |    |

$a_1a_2 = 00$

$a_5a_6$

|             |    |    |    |    |
|-------------|----|----|----|----|
|             | 00 | 01 | 11 | 10 |
| $a_3a_4$ 00 |    |    |    |    |
| 01          |    |    |    |    |
| 11          |    |    |    |    |
| 10          |    |    |    |    |

$a_1a_2 = 01$

$a_5a_6$

|             |    |    |    |    |
|-------------|----|----|----|----|
|             | 00 | 01 | 11 | 10 |
| $a_3a_4$ 00 |    | 1  | 1  | 1  |
| 01          | 1  | 1  | 1  | 1  |
| 11          | 1  | 1  | 1  | 1  |
| 10          | 1  | 1  | 1  | 1  |

$a_1a_2 = 10$

$a_5a_6$

|             |    |    |    |    |
|-------------|----|----|----|----|
|             | 00 | 01 | 11 | 10 |
| $a_3a_4$ 00 | 1  | 1  | 1  | 1  |
| 01          | 1  | 1  | 1  | 1  |
| 11          |    |    |    |    |
| 10          | 1  | 1  |    |    |

$a_1a_2 = 11$

$$C_1 = a_4a_5$$

1XX0X

$a_5a_6$

|             |    |    |    |    |
|-------------|----|----|----|----|
|             | 00 | 01 | 11 | 10 |
| $a_3a_4$ 00 |    |    |    |    |
| 01          |    |    |    |    |
| 11          |    |    |    |    |
| 10          |    |    |    |    |

$a_1a_2 = 00$

$a_5a_6$

|             |    |    |    |    |
|-------------|----|----|----|----|
|             | 00 | 01 | 11 | 10 |
| $a_3a_4$ 00 |    | 1  | 1  | 1  |
| 01          | 1  | 1  | 1  | 1  |
| 11          | 1  | 1  | 1  | 1  |
| 10          | 1  | 1  | 1  | 1  |

$a_1a_2 = 01$

$a_5a_6$

|             |    |    |    |    |
|-------------|----|----|----|----|
|             | 00 | 01 | 11 | 10 |
| $a_3a_4$ 00 | 1  |    |    |    |
| 01          |    |    |    |    |
| 11          |    |    |    |    |
| 10          |    |    |    |    |

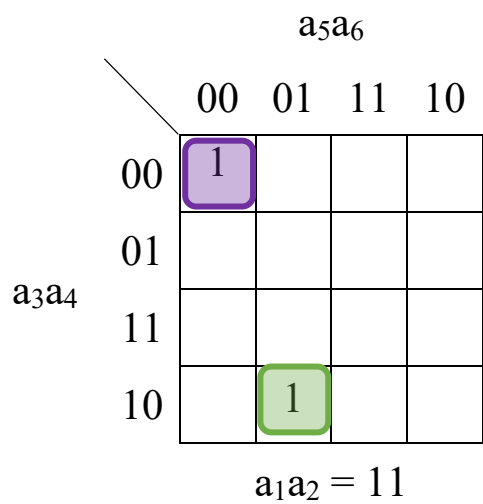
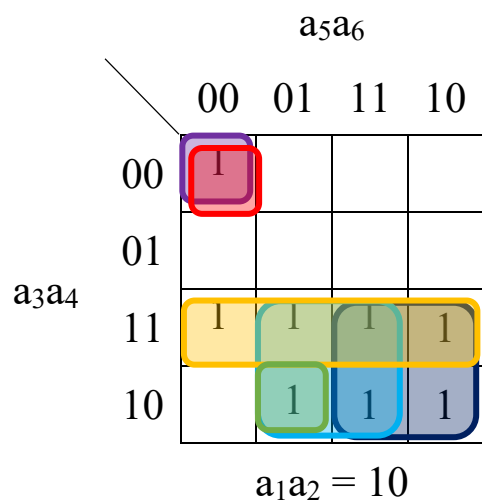
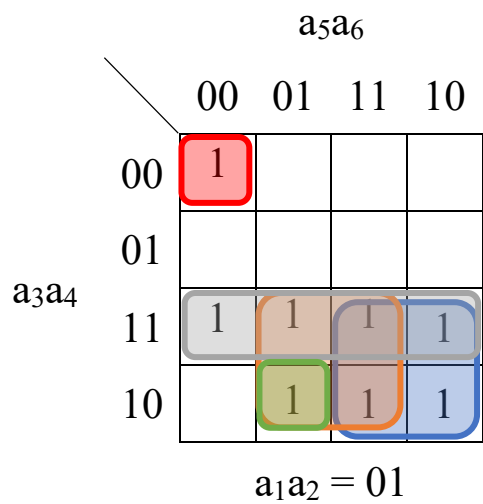
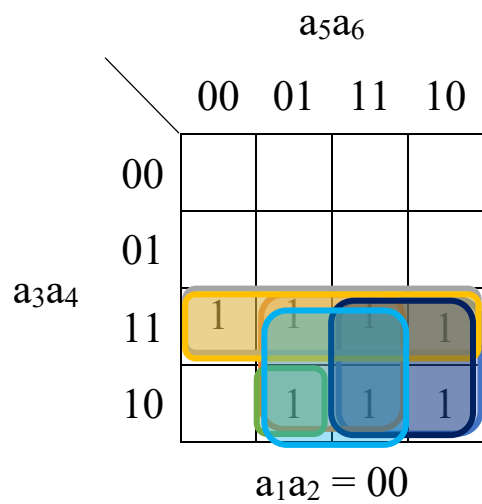
$a_1a_2 = 10$

$a_5a_6$

|             |    |    |    |    |
|-------------|----|----|----|----|
|             | 00 | 01 | 11 | 10 |
| $a_3a_4$ 00 |    | 1  | 1  | 1  |
| 01          | 1  | 1  | 1  | 1  |
| 11          |    |    |    |    |
| 10          | 1  | 1  |    |    |

$a_1a_2 = 11$

C2



C3

$a_5a_6$

|    |    |    |    |    |
|----|----|----|----|----|
|    | 00 | 01 | 11 | 10 |
| 00 | 1  |    |    |    |
| 01 |    | 1  | 1  |    |
| 11 |    | 1  | 1  |    |
| 10 | 1  |    |    |    |

$a_1a_2 = 00$

$a_5a_6$

|    |    |    |    |    |
|----|----|----|----|----|
|    | 00 | 01 | 11 | 10 |
| 00 | 1  |    |    |    |
| 01 |    | 1  | 1  |    |
| 11 |    | 1  | 1  |    |
| 10 | 1  |    |    |    |

$a_1a_2 = 01$

$a_5a_6$

|    |    |    |    |    |
|----|----|----|----|----|
|    | 00 | 01 | 11 | 10 |
| 00 | 1  |    |    |    |
| 01 |    | 1  | 1  |    |
| 11 |    | 1  | 1  |    |
| 10 | 1  |    |    |    |

$a_1a_2 = 10$

$a_5a_6$

|    |    |    |    |    |
|----|----|----|----|----|
|    | 00 | 01 | 11 | 10 |
| 00 | 1  |    |    |    |
| 01 |    | 1  | 1  |    |
| 11 |    |    | 1  | 1  |
| 10 | 1  |    |    |    |

$a_1a_2 = 11$

C4

|          |    |               |    |    |    |
|----------|----|---------------|----|----|----|
|          |    | $a_5a_6$      |    |    |    |
|          |    | 00            | 01 | 11 | 10 |
| $a_3a_4$ | 00 |               |    | 1  |    |
|          | 01 | 1             |    | 1  |    |
|          | 11 | 1             |    | 1  |    |
|          | 10 | 1             |    | 1  |    |
|          |    | $a_1a_2 = 00$ |    |    |    |

|          |    |               |    |    |    |
|----------|----|---------------|----|----|----|
|          |    | $a_5a_6$      |    |    |    |
|          |    | 00            | 01 | 11 | 10 |
| $a_3a_4$ | 00 | 1             |    | 1  |    |
|          | 01 | 1             |    | 1  |    |
|          | 11 | 1             |    | 1  |    |
|          | 10 | 1             |    | 1  |    |
|          |    | $a_1a_2 = 01$ |    |    |    |

|          |    |               |    |    |    |
|----------|----|---------------|----|----|----|
|          |    | $a_5a_6$      |    |    |    |
|          |    | 00            | 01 | 11 | 10 |
| $a_3a_4$ | 00 | 1             |    | 1  |    |
|          | 01 | 1             |    | 1  |    |
|          | 11 | 1             |    | 1  |    |
|          | 10 | 1             |    | 1  |    |
|          |    | $a_1a_2 = 10$ |    |    |    |

|          |    |               |    |    |    |
|----------|----|---------------|----|----|----|
|          |    | $a_5a_6$      |    |    |    |
|          |    | 00            | 01 | 11 | 10 |
| $a_3a_4$ | 00 | 1             |    | 1  |    |
|          | 01 | 1             |    | 1  |    |
|          | 11 | 1             | 1  |    |    |
|          | 10 | 1             |    |    |    |
|          |    | $a_1a_2 = 11$ |    |    |    |

C5

|          |    |               |    |    |    |
|----------|----|---------------|----|----|----|
|          |    | $a_5a_6$      |    |    |    |
|          |    | 00            | 01 | 11 | 10 |
| $a_3a_4$ | 00 | 1             |    |    | 1  |
|          | 01 | 1             |    |    | 1  |
|          | 11 | 1             |    |    | 1  |
|          | 10 | 1             |    |    | 1  |
|          |    | $a_1a_2 = 00$ |    |    |    |

|          |    |               |    |    |    |
|----------|----|---------------|----|----|----|
|          |    | $a_5a_6$      |    |    |    |
|          |    | 00            | 01 | 11 | 10 |
| $a_3a_4$ | 00 | 1             |    |    | 1  |
|          | 01 | 1             |    |    | 1  |
|          | 11 | 1             |    |    | 1  |
|          | 10 | 1             |    |    | 1  |
|          |    | $a_1a_2 = 01$ |    |    |    |

|          |    |               |    |    |    |
|----------|----|---------------|----|----|----|
|          |    | $a_5a_6$      |    |    |    |
|          |    | 00            | 01 | 11 | 10 |
| $a_3a_4$ | 00 | 1             |    |    | 1  |
|          | 01 | 1             |    |    | 1  |
|          | 11 | 1             |    |    | 1  |
|          | 10 | 1             |    |    | 1  |
|          |    | $a_1a_2 = 10$ |    |    |    |

|          |    |               |    |    |    |
|----------|----|---------------|----|----|----|
|          |    | $a_5a_6$      |    |    |    |
|          |    | 00            | 01 | 11 | 10 |
| $a_3a_4$ | 00 | 1             |    |    | 1  |
|          | 01 | 1             |    |    | 1  |
|          | 11 |               | 1  | 1  |    |
|          | 10 | 1             |    | 1  |    |
|          |    | $a_1a_2 = 11$ |    |    |    |

C6